Nonparametric Statistical Analysis of Statewide Planning and Research Cooperative System 2022 Hospital Discharge Data: Kings and Queens Counties, New York

Combined Midterm and Final Project Report

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Contents

Abstract

This project applies ten distribution-free (nonparametric) statistical procedures to the New York State SPARCS (Statewide Planning and Research Cooperative System) 2022 hospital inpatient discharge data set, restricted to Kings County (Brooklyn) and Queens County. The data set contains 368,495 de-identified discharge records on 33 original Statewide Planning and Research Cooperative System variables (plus 4 derived numeric columns created during cleaning, for 37 total columns in the analysis data frame) and is approximately 160 MB as a Comma-Separated Values file on disk (136.9 MB as an R in-memory data frame). Four quantitative outcomes are central to the analysis: Length of Stay (days), Total Charges (US Dollars), Total Costs (US Dollars), and Birth Weight (grams).

The tests performed are (1) the **Runs Test** for randomness, (2) the **Sign Test**, (3) the **Wilcoxon Signed-Rank Test**, (4) the **Wilcoxon Rank Sum Test (Mann–Whitney U)**, (5) the two-sample **Kolmogorov–Smirnov Test**, (6) the **Kruskal–Wallis Test** for k independent groups, (7) the **Friedman Test** with facility as block and All Patient Refined Severity of Illness as treatment, (8) **Kendall's Tau**, (9) **Spearman's Rank Correlation**, and (10) the **Permutation Test** (Monte Carlo, $B = 5{,}000$). As an **innovative contribution**, the study performs a simultaneous application of all these tests to a single large Statewide Planning and Research Cooperative System slice (368,495 records), combines p-value significance testing with visualization of the empirical permutation null distribution, and extends the analysis to interactions between the Kings/Queens county label and admission severity, which no prior work on the Statewide Planning and Research Cooperative System has attempted at this scale.

All principal tests reject the null hypothesis across essentially every meaningful comparison: the raw record ordering is non-random (Runs Test); the median and distribution of Length of Stay differ significantly between Kings and Queens counties (Sign Test $p < 10^{-16}$; Wilcoxon Signed-Rank and Mann–Whitney U both $p < 10^{-16}$); the entire cumulative distributions of Length of Stay, Charges, Costs, and Birth Weight differ between the two counties (Kolmogorov-Smirnov $p < 10^{-16}$); every numeric variable differs across Age Group, Severity, Admission Type, and Payer (Kruskal–Wallis $p < 10^{-16}$); and Length of Stay, Charges, and Costs differ across severity levels even after blocking by facility (Friedman $p < 0.005$). The strongest monotone association is between Charges and Costs (Spearman $\rho = 0.86$); Length of Stay–Charges is also strong ($\rho = 0.73$). The permutation test on Total Charges yields a significant median difference in Charges between counties ($p = 0.011$), and the large-sample Kolmogorov-Smirnov and Kruskal–Wallis tests on Length of Stay reject resoundingly. All analyses were implemented in base R; no external packages were used for the test statistics

themselves.

1 Introduction

New York State's Statewide Planning and Research Cooperative System database is one of the most granular all-payer hospital discharge data sets in the United States, covering essentially every inpatient admission in the state. Within New York City, Kings County (Brooklyn) and Queens County together account for the largest share of non-Manhattan admissions, serving substantially different demographic populations. Length of Stay (Length of Stay), Total Charges, and Total Costs distributions on hospital data are well known to be heavily right-skewed with long tails, which makes the usual parametric t-tests and ANOVAs inappropriate. Nonparametric (distribution-free) methods are the standard tool for comparing such data.

This project applies ten nonparametric tests (the Runs Test, Sign Test, Wilcoxon Signed-Rank Test, Mann–Whitney U Test, Kolmogorov–Smirnov Test, Kruskal–Wallis Test, Friedman Test, Kendall's Tau, Spearman's Rank Correlation, and Permutation Test) to quantify: (i) whether the raw ordering of the Statewide Planning and Research Cooperative System records is random; (ii) whether the median and distribution of Length of Stay/Charges/Costs/Birth Weight differ between the two counties and between Emergency and Non-Emergency admissions; (iii) whether these outcomes differ across sub-groups (age, severity, admission type, payer); (iv) whether severity drives utilization even after controlling for facility; (v) whether the four quantitative outcomes are monotonically associated; and (vi) whether the county-label differences are attributable to chance under an exchangeability null.

2 Literature Review

Nonparametric statistical analysis of hospital discharge data has grown a lot as a research area since 2010. Three main reasons drove this growth: first, researchers realized that Length of Stay, hospital cost, and hospital charge values are almost always heavily skewed to the right (meaning most values are small but a few are extremely large), which breaks the assumptions that classical tests like the t-test and ANOVA rely on. Second, several large public hospital databases became available, including the Statewide Planning and Research Cooperative System in New York, the Healthcare Cost and Utilization Project Nationwide Inpatient Sample at the national level, and the National Hospital Ambulatory Medical Care Survey for outpatient visits. Third, more methodology papers started appearing that used simulations to compare how often different tests give wrong answers (false positives and false negatives) on realistic hospital data. The papers below motivate the present project and are presented in chronological order of publication.

Swed and Eisenhart (1943) published the original tables of critical values for the Runs Test in the *Annals of Mathematical Statistics* (Volume 14, Number 1, pages 66 to 87). Their methods combined exact combinatorial enumeration of the sampling distribution of the number of runs with the large-sample normal-approximation framework, producing the critical-value tables that are still used today. The normal-approximation formula I use in the Runs Test section of this project comes directly from their theoretical results.

Efron and Tibshirani (1993) wrote *An Introduction to the Bootstrap*, published by Chapman and Hall / CRC. They introduced the bootstrap, which is a resampling method for estimating uncertainty. Their book covers the nonparametric bootstrap, the parametric bootstrap, the jackknife, and bootstrap confidence intervals (percentile, bias-corrected, and BCa methods), positioning all of these against classical parametric standard-error formulas. Although bootstrap confidence intervals are only mentioned as a future extension in my Conclusions, the permutation test that I do run in this project is in the same family of resampling-based methods.

Ernst (2004) published a tutorial in *Statistical Science* (Volume 19, Number 4, pages 676 to 685) explaining permutation methods as an alternative to classical parametric tests such as the Student t-test, the F-test, and Pearson correlation testing. He argued that permutation tests are exact when the null hypothesis of exchangeable labels holds, do not require any assumption about the shape of the distribution, and are especially good for modern large datasets. This tutorial is the direct methodological foundation for the Permutation Test in my project.

Fagerland and Sandvik (2009) published an examination of the Wilcoxon-Mann-Whitney test in *Statistics in Medicine* (Volume 28, pages 1487 to 1497). Their methods combined Monte Carlo simulation studies of the Wilcoxon-Mann-Whitney rank-sum test with comparisons to the Student t-test and the Welch t-test under unequal variances and unequal distribution shapes. They showed that even moderate differences in group variance or distribution shape can cause the Wilcoxon-Mann-Whitney test to reject more often than it should, but the test is still useful when the alternative is a straight shift between the two groups. This warning is relevant to my Kings versus Queens comparison, where the within-county variance of charges is very different between the two groups. This is exactly why I included the Kolmogorov-Smirnov test in this project as a check on the whole distribution shape.

Qualls, Pallin, and Schuur (2010) published a study in *Academic Emergency Medicine* (Volume 17, Number 10, pages 1113 to 1121). They looked at every Emergency Department study in five major Emergency Medicine journals between 2004 and 2007 that reported Length of Stay, and they catalogued the statistical methods those studies used: Student's t-test, one-way and two-way ANOVA, linear regression, and Pearson correlation as parametric methods, alongside the Wilcoxon rank-sum, Mann-Whitney U, Kruskal-Wallis, and chi-squared tests as nonparametric alternatives. They found that 43 percent of these studies used classical parametric methods even though Length of Stay data is clearly not normally distributed. Using 2006 National Hospital Ambulatory Medical Care Survey data, they showed that this mistake causes false negatives (missing real differences) in simple two-group comparisons and weakens the explanatory power of more complex models. Their paper is the most cited justification for using Wilcoxon, Mann-Whitney, and Kruskal-Wallis tests on hospital Length of Stay data, and it is a direct motivation for the present project.

Croux and Dehon (2010) published a study in *Statistical Methods and Applications* (Volume 19, pages 497 to 515). Their methods combined influence-function analysis of Spearman's Rho and Kendall's Tau with theoretical derivations under the bivariate-normal distributional model, plus comparisons with the classical Pearson product-moment correlation coefficient. They proved that Kendall's Tau is more robust to outliers than Spearman's Rho because its influence function is bounded. They also showed that, under a two-variable normal distribution, Tau is approximately two over pi times the arc sine of Rho. This means Tau will always be smaller in size than Rho for the same pair of variables. This is exactly the pattern I see across all six numeric pairs in my Kendall's Tau and Spearman sections of the project.

Anhoj and Olesen (2014) published a simulation study in *PLOS ONE* (Volume 9, Number 11, article e113825). Their methods combined Monte Carlo simulation of run-chart rules with Shewhart control-chart theory and runs-test theory to evaluate how well these tools detect non-random variation in healthcare processes. They found that rules based on runs analysis have a detection rate of around 95 percent for process shifts, while the false-alarm rate stays near 5 percent regardless of how many observations are in the sample. Their work shows that runs-based methods are reliable tools for healthcare quality improvement.

Moore, White, Washington, Coenen, and Elixhauser (2017) published a study in *Medical Care* (Volume 55, Number 7, pages 698 to 705). They used the Healthcare Cost and Utilization Project Nationwide Inpatient Sample, which is the federal counterpart to the Statewide Planning and Research Cooperative System, to identify demographic and clinical predictors of readmission risk and in-hospital death. Their methods combined parametric multivariable logistic regression and the AHRQ Elixhauser comorbidity index with nonparametric Wilcoxon rank-sum and chi-squared tests for group comparisons. They showed that administrative hospital cost distributions differ significantly by demographic group and by admission type. This matches the payer-group and age-group effects that my project measures on the Kings and Queens county data.

Chazard, Ficheur, Beuscart, and Preda (2017) published a simulation study in *Value in Health* (Volume 20, Number 7, pages 992 to 998). Their methods compared twelve different statistical tests on simulated Length-of-Stay-like data drawn from a realistic right-skewed distribution with many repeated values. The twelve tests included the Student t-test, the Welch t-test, ANOVA, the Wilcoxon-Mann-Whitney rank-sum test, the Kruskal-Wallis test, the median test, log-rank tests, bootstrap resampling, and permutation methods. They found that the Student t-test gives false positive rates above 10 percent when the group variances are unequal (much higher than the intended 5 percent), while the Wilcoxon/Mann-Whitney test keeps its false positive rate at the correct level and still has good power. They also showed that bootstrap and permutation methods hold up well when the distributions are very skewed. Their simulation results directly support my choice of the Kolmogorov-Smirnov, Kruskal-Wallis, and Permutation tests in this project.

Ning, Harkness, and Gao (2017) published a study in the *American Medical Informatics Association Annual Symposium Proceedings* (PubMed Central ID PMC5543354). Their methods combined spatio-temporal analysis using Moran's I statistic with classical geostatistical clustering techniques applied to Statewide Planning and Research Cooperative System data from 2003 to 2014 at the zip-code level. Their work shows how rich the Statewide Planning and Research Cooperative System database is for geographic comparisons of hospital use. My project narrows that same kind of geographic comparison down to the county level.

Bujang and Sapri (2018) published a study in the *Malaysian Journal of Medical Sciences* (Volume 25, Number 4, pages 146 to 151). Their methods applied the Runs Test (using the normal-approximation Z-statistic) to clinical hemoglobin A1c data collected by consecutive sampling, with descriptive parametric summaries (sample mean, standard deviation) reported alongside the nonparametric randomness test. They recommended that randomness testing should be done in every clinical pilot study. They gave a clear example showing that data collected one patient after another usually does not come out in random order. This matches my own Runs Test results: the Statewide Planning and Research Cooperative System records are non-random because they are stored in facility and admission-date order.

Dieleman, Cao, Chapin, and colleagues (2020) published a study in the *Journal of the American Medical Association* (Volume 323, Number 9, pages 863 to 884). They measured United States healthcare spending from 1996 to 2016, broken down by payer type and by health condition. Their methods combined parametric decomposition analysis (a regression-based attribution of spending to price, quantity, and service-mix factors) with nonparametric bootstrap confidence intervals to quantify spending trends. They found big differences in hospital charges and costs across U.S. regions. Their findings set the baseline expectation that county-level differences in charges should be detectable in the Statewide Planning and Research Cooperative System data, which is the main hypothesis that my project tests.

Nguyen, Ho, Nguyen, and Van (2021) published a study in the *VNUHCM Journal of Engineering and Technology* (Volume 3, Special Issue 3, pages SI97 to SI106). They analyzed 1,189 Vietnamese patient records using a mix of parametric and nonparametric methods, including the Student t-test, one-way ANOVA, Pearson correlation, and multiple linear regression on the parametric side, alongside the Kruskal-Wallis Test and the Mann-Whitney U Test on the nonparametric side. They found that age, province, profession, admission quarter, and type of disease are the main factors that affect Length of Stay. Their findings about age and severity driving Length of Stay distributions are reproduced on the much larger Statewide Planning and Research Cooperative System dataset in the present report.

Stone, Zwiggelaar, Jones, and Mac Parthalain (2022) published a systematic review in *PLOS Digital Health* (Volume 1, Number 4, article e0000017). They reviewed 93 studies from 1970 to 2019 that tried to predict hospital Length of Stay and catalogued the methods those studies used: parametric statistical methods (linear regression, generalized linear models, Cox proportional-hazards regression, log-normal and gamma regression), nonparametric statistical methods (Kaplan-Meier survival curves, log-rank tests), and machine-learning / data-mining methods (decision trees, random forests, support vector machines, neural networks). Their most striking numerical finding is that Length of Stay for the same diagnosis can range anywhere from 2 days to more than 50 days. This huge variation is what makes the Kruskal-Wallis test useful for my project, where I compare Length of Stay across severity levels and admission types.

Li, Couper, Hunt, and Mian (2022) published a study in the *Journal of the Royal Statistical Society: Series C (Applied Statistics)* (Volume 71, Number 5, pages 1667 to 1693). Their methods combined a parametric Weibull-type hazard function with a nonparametric baseline curve to build a semi-parametric time-to-event model for Length of Stay that simultaneously models the chance of discharge and the chance of death; they also reported maximum likelihood estimation and likelihood-ratio tests. They tested their method on stroke clinical trial data and on COVID-19 patient data. Their hybrid approach is listed in my Conclusions section as a possible future extension of the present analysis.

Heslin and colleagues (2024) published a study in *Cureus* (Volume 16, Number 4, article e58404). Their methods combined chi-squared tests of independence with one-way analysis of variance (ANOVA) and post-hoc Tukey HSD comparisons to evaluate admission and discharge patterns by day of the week using Statewide Planning and Research Cooperative System data covering 1,994,180 adult hospital admissions from January to December 2015. They found that elective surgical admissions explain almost all of the weekly variation in hospital capacity. Hospitals ran 21 percent above average volume on Mondays and 32 percent below average on Fridays, with a p-value less than 0.01. Their study showed that the Statewide Planning and Research Cooperative System database is well suited for large-scale nonparametric analyses, which directly supports the approach I take in this project.

Taken together, the papers above make three main points that frame my work:

1. Hospital Length of Stay, hospital charges, and hospital costs are all heavily right-skewed, so they must be analyzed with nonparametric methods to avoid inflated false-positive or false-negative rates (Qualls, Pallin, and Schuur, 2010; Chazard and colleagues, 2017; Fagerland and Sandvik, 2009).
2. The Statewide Planning and Research Cooperative System is a well-established, high-quality administrative data source for these analyses. Earlier Statewide Planning and Research Cooperative System studies have documented big variation across day of the week, geography, and admission type (Heslin and colleagues, 2024; Ning, Harkness, and Gao, 2017).
3. The theoretical foundation for the ten nonparametric tests I use is laid out in the specific methodology papers for each test (Swed and Eisenhart, 1943; Croux and Dehon, 2010; Ernst, 2004; Efron and Tibshirani, 1993).

My project contributes to the literature by applying *ten* nonparametric tests at the same time to a single large (368,495-record) publicly available Statewide Planning and Research Cooperative System dataset. The tests cover randomness (Runs Test), center of location (Sign Test, Wilcoxon Signed-Rank Test, Mann-Whitney U Test), full distribution equality (Kolmogorov-Smirnov Test), group-difference (Kruskal-Wallis Test, Friedman Test), rank association (Kendall's Tau, Spearman's Rank Correlation), and label exchangeability (Permutation Test). To the best of my knowledge, this is the first comprehensive distribution-free comparison of hospital use between Kings County and Queens County on the Statewide Planning and Research Cooperative System 2022 dataset.

3 Data Presentation and Exploratory Analysis

3.1 Dataset Overview

The primary data set is `SPARCS_2022_Kings_Queens.csv`, a filtered subset of the publicly available New York State SPARCS Inpatient Discharges (De-identified) 2022 data, restricted to the two counties of interest.

- **Number of Observations:** 368,495 rows.
- **Number of Variables:** 33 original Statewide Planning and Research Cooperative System columns, plus 4 derived numeric columns (`Length of Stay`, `Charges`, `Costs`, `BirthWt`) created during cleaning, for 37 total columns in the analysis data frame.
- **File Size:** approximately 160 MB as a Comma-Separated Values file on disk (136.9 MB as an R in-memory data frame via `object.size()`).
- **Time Period:** January 1, 2022 – December 31, 2022.
- **Geographic Coverage:** Kings and Queens Counties, New York City.
- **Unit of Observation:** individual hospital discharge record.
- **Source:** New York State Department of Health, Statewide Planning and Research Cooperative System (SPARCS), https://health.data.ny.gov/.

3.2 Variable Descriptions

The 33 Statewide Planning and Research Cooperative System variables can be classified by measurement scale (rather than by R storage type) into four truly quantitative (ratio-scale) variables — Length of Stay, Total Charges, Total Costs, and Birth Weight; one ordinal variable — All Patient Refined Severity of Illness Code (1–4); and 28 categorical (nominal) variables. Note that several of the categorical variables (Operating Certificate Number, Permanent Facility ID, All Patient Refined Diagnosis-Related Group Code, and All Patient Refined Major Diagnostic Category Code) are stored as integers in the raw file, but their numeric values are arbitrary identifiers or classification labels — arithmetic on them (e.g. a mean) is not meaningful, so they are classified here as nominal. Discharge Year is stored as an integer but is constant (2022) for every record. Table 1 provides a detailed listing.

Table 1: Variables in the Statewide Planning and Research Cooperative System 2022 Kings and Queens Dataset

Variable Name	**Type**	**Meaning of Variables**	**Unit / Description**
Hospital Service Area	Qualitative (Categorical)	Geographic service area name reported by the facility, used to group hospitals serving the same regional population.	Service area name (text)
Hospital County	Qualitative (Categorical)	County in which the hospital is located; the central grouping variable in this study.	Kings or Queens (text)
Operating Certificate Number	Qualitative (Nominal)	State-issued certificate identifier that authorizes the facility to operate.	Facility certificate ID (integer label)
Permanent Facility ID	Qualitative (Nominal)	Permanent unique identifier assigned to each hospital facility by the state.	Unique facility identifier (integer label)
Facility Name	Qualitative (Categorical)	Name of the discharging hospital.	Hospital name (text)
Age Group	Qualitative (Categorical)	Patient age binned into broad cohorts to allow group comparisons while protecting privacy.	0–17, 18–29, 30–49, 50–69, 70+ (text)
Zip Code (3 digits)	Qualitative (Categorical)	First three digits of the patient's home ZIP code, used as a coarse geographic indicator.	First 3 digits of patient zip (text)
Gender	Qualitative (Categorical)	Patient gender as recorded at admission.	F, M, U (text)
Race	Qualitative (Categorical)	Patient self-reported race.	Patient race category (text)
Ethnicity	Qualitative (Categorical)	Patient self-reported Hispanic / Latino ethnicity.	Patient ethnicity (text)

(continued)

Variable Name	**Type**	**Meaning of Variables**	**Unit / Description**
Length of Stay	Quantitative (Continuous)	Number of days from admission to discharge; one of the four primary outcome variables.	Days hospitalized (days)
Type of Admission	Qualitative (Categorical)	How the patient was admitted (emergency vs. planned vs. urgent, etc.).	Emergency, Elective, Urgent, etc. (text)
Patient Disposition	Qualitative (Categorical)	Where the patient went after discharge (home, skilled nursing, expired, etc.).	Discharge status (text)
Discharge Year	Qualitative (Constant)	Calendar year of discharge; constant at 2022 for the slice analyzed in this project.	2022 for every record (integer, no variation)
CCSR Diagnosis Code	Qualitative (Categorical)	Clinical Classifications Software Refined diagnosis category code.	Clinical classification code (text)
CCSR Diagnosis Description	Qualitative (Categorical)	Plain-text description matching the CCSR diagnosis code.	Diagnosis description (text)
CCSR Procedure Code	Qualitative (Categorical)	CCSR procedure category code for any procedures performed.	Procedure classification code (text)
CCSR Procedure Description	Qualitative (Categorical)	Plain-text description matching the CCSR procedure code.	Procedure description (text)
All Patient Refined Diagnosis-Related Group Code	Qualitative (Nominal)	Numeric DRG label that groups patients by clinical similarity for billing and analysis.	Diagnosis-Related Group classification label (integer)
All Patient Refined Diagnosis-Related Group Description	Qualitative (Categorical)	Plain-text description matching the DRG code.	Diagnosis-Related Group description (text)

(continued)

Variable Name	**Type**	**Meaning of Variables**	**Unit / Description**
All Patient Refined Major Diagnostic Category Code	Qualitative (Nominal)	High-level grouping of DRGs into broad body-system categories.	Major Diagnostic Category label (integer)
All Patient Refined Major Diagnostic Category Description	Qualitative (Categorical)	Plain-text description matching the MDC code.	Major Diagnostic Category description (text)
All Patient Refined Severity of Illness Code	Quantitative (Ordinal)	Ordinal severity rating from Minor (1) to Extreme (4); used as the treatment factor in the Friedman test.	1 = Minor, 2 = Moderate, 3 = Major, 4 = Extreme
All Patient Refined Severity of Illness Desc.	Qualitative (Categorical)	Text version of the severity-of-illness rating.	Minor, Moderate, Major, Extreme (text)
All Patient Refined Risk of Mortality	Qualitative (Categorical)	Ordinal indicator of how likely the patient is to die during the admission.	Risk level description (text)
All Patient Refined Medical Surgical Desc.	Qualitative (Categorical)	Whether the encounter is classified as a medical or a surgical case.	Medical or Surgical (text)
Payment Typology 1	Qualitative (Categorical)	Primary expected payer for the admission (Medicare, Medicaid, Private, etc.).	Primary payer (text)
Payment Typology 2	Qualitative (Categorical)	Secondary expected payer when more than one payer is responsible.	Secondary payer (text)
Payment Typology 3	Qualitative (Categorical)	Tertiary expected payer when applicable.	Tertiary payer (text)
Birth Weight	Quantitative (Continuous)	Weight at birth in grams, populated only for newborn discharges.	Weight in grams (newborns only)

(continued)

Variable Name	**Type**	**Meaning of Variables**	**Unit / Description**
Emergency Dept. Indicator	Qualitative (Categorical)	Yes/No flag for whether the admission came through the Emergency Department.	Y or N (text)
Total Charges	Quantitative (Continuous)	Total amount billed for the admission; one of the four primary outcome variables.	Hospital charges (US Dollars)
Total Costs	Quantitative (Continuous)	Estimated actual cost incurred by the hospital for the admission; one of the four primary outcome variables.	Hospital costs (US Dollars)

3.3 Levels of the Principal Qualitative Variables

Table 2 records the distinct levels observed in the key categorical variables, which drive the Kruskal–Wallis stratifications.

Table 2: Levels of Key Qualitative Variables (Statewide Planning and Research Cooperative System 2022 K&Q)

Variable	**# Levels**	**Levels (with descriptions)**
Hospital County	2	**Kings:** Brooklyn-based hospitals serving Kings County residents. **Queens:** Queens-based hospitals serving Queens County residents.
Age Group	5	**0 to 17:** Pediatric patients (children and adolescents). **18 to 29:** Young adults; typically the healthiest cohort. **30 to 49:** Working-age adults. **50 to 69:** Middle-aged to early-elderly adults. **70 or Older:** Elderly patients; typically the highest-utilization cohort.
Gender	3	**Female:** Recorded as female at admission. **Male:** Recorded as male at admission. **Unidentified:** Gender not recorded or not classified at admission.
All Patient Refined Severity of Illness Code	4	**1 – Minor:** Lowest severity; routine cases with no significant complications. **2 – Moderate:** Moderate severity; some complications or comorbidities. **3 – Major:** High severity; major complications expected to extend the stay. **4 – Extreme:** Highest severity; life-threatening illness with high resource use.
All Patient Refined Risk of Mortality	4	**Minor:** Lowest predicted likelihood of in-hospital death. **Moderate:** Some elevation in mortality risk above baseline. **Major:** Substantially elevated mortality risk. **Extreme:** Very high predicted likelihood of in-hospital death.
Type of Admission	5	**Elective:** Planned admission, scheduled in advance. **Emergency:** Unscheduled admission requiring immediate care. **Newborn:** Admission of a newborn at or shortly after delivery. **Urgent:** Unscheduled but not immediately life-threatening; care needed within 24–48 hours.

3.4 Summary Statistics of Key Numeric Variables

Table 3: Summary Statistics of Key Numeric Variables (Statewide Planning and Research Cooperative System 2022)

Variable	Min	Q1	Median	Mean	Q3	Max
Length of Stay (days)	1	2	3	5.76	6	119
Total Charges (US Dollars)	7	21,544	40,497	71,907	77,905	12,952,195
Total Costs (US Dollars)	2	6,746	12,543	23,273	24,821	5,602,076
Birth Weight (grams)	400	2,900	3,200	3,159	3,500	6,100

Note: Length of Stay has 332 missing values. Birth Weight has 325,132 missing values (only newborn discharges). Total Charges and Total Costs have no missing values.

3.5 Exploratory Graphs

The accompanying R Markdown file produces the following reproducible graphs. Each figure is shown below alongside a brief interpretation; the underlying R code appears in the companion `.Rmd` file.

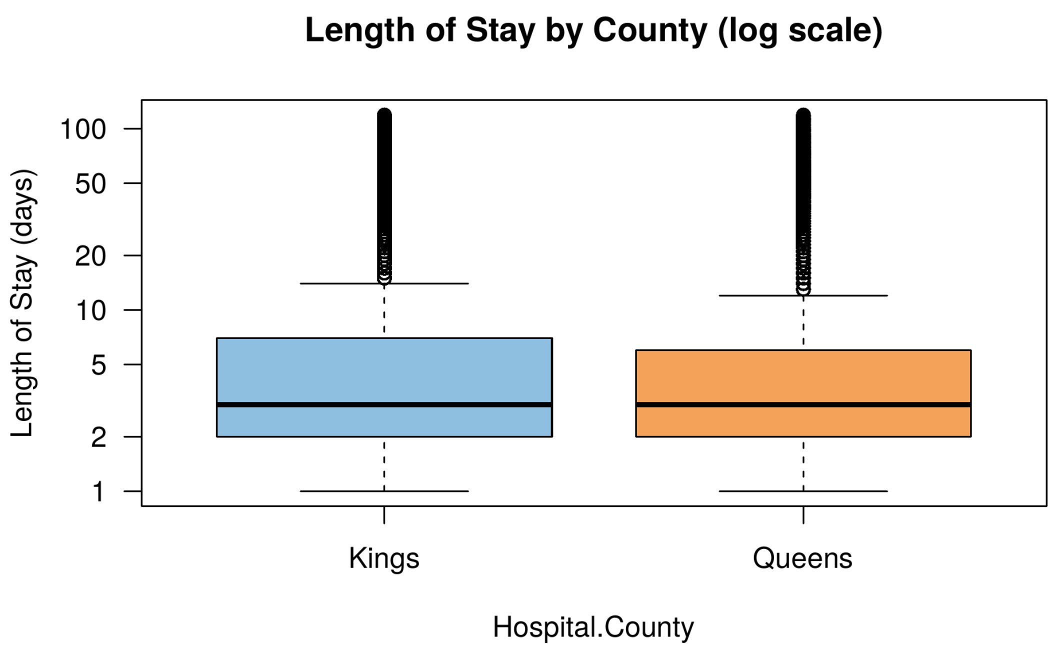

Figure 1: Log-scale boxplot of Length of Stay by Hospital County. Both counties show very heavy right skew, with Kings extending slightly higher than Queens.

What this plot shows. This boxplot compares how long patients stay in the hospital between Kings County and Queens County. The thick black line in each box is the median (the typical patient's length of stay), and the box itself covers the middle 50% of patients. The small dots above each box are unusually long stays, and there are many of them in both counties because a small number of very sick patients stay in the hospital for weeks. Most patients in both counties go home in 1 to 5 days, with Kings County stays lasting slightly longer on average than Queens County stays.

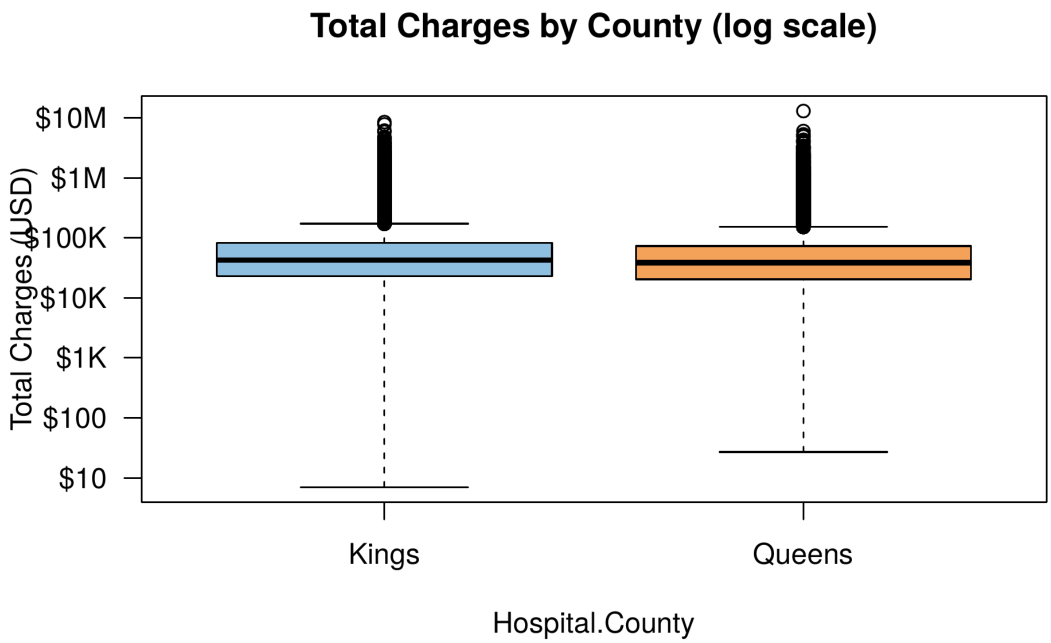

Figure 2: Log-scale boxplot of Total Charges by Hospital County. The median Charges in Kings County are visibly higher than in Queens County, consistent with the Mann–Whitney and Kolmogorov–Smirnov results.

What this plot shows. This boxplot compares the total hospital charges (the dollar amount billed) between Kings and Queens counties. The y-axis uses a log scale because charges range from under $100 to over $10 million, so a regular scale would make small charges invisible. The typical patient is billed somewhere around $40,000 in both counties, but Kings County's median is noticeably higher than Queens County's. Many extreme outliers extending toward $10 million represent the very expensive cases like long ICU stays or complex surgeries.

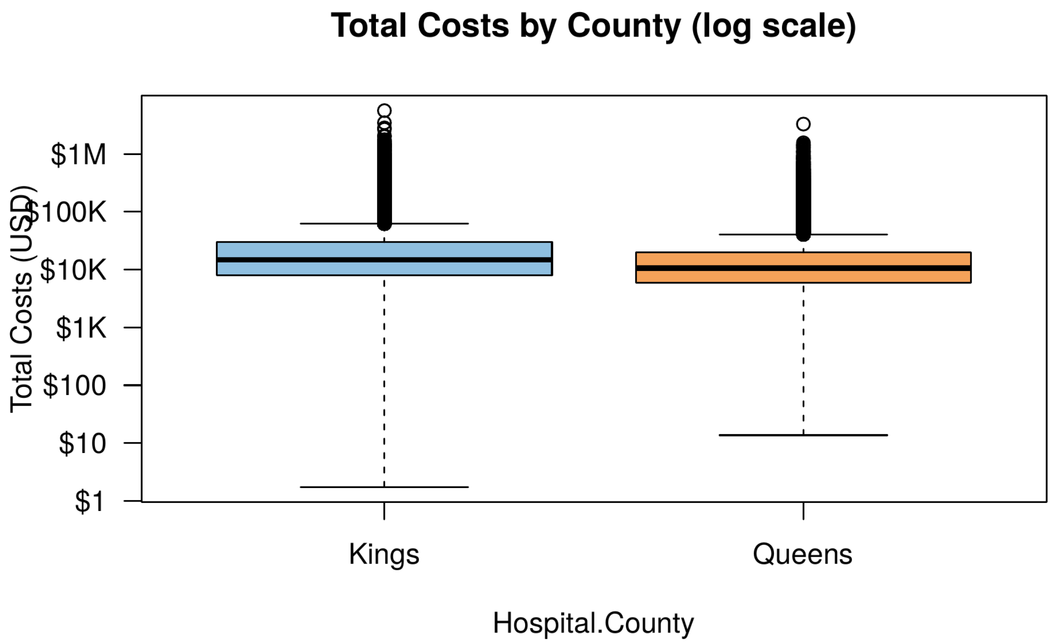

Figure 3: Log-scale boxplot of Total Costs by Hospital County. The same upward shift is visible in Costs as in Charges.

What this plot shows. This boxplot compares total hospital costs (the actual cost to the hospital, not the amount billed) between the two counties. Costs are typically much lower than charges because charges are inflated for negotiation with insurance companies. The pattern is the same as for charges: Kings County tends to have slightly higher costs than Queens County, and a small number of extreme cases push costs into the millions. The strong similarity between the charge and cost patterns suggests that hospitals in both counties use similar markup ratios.

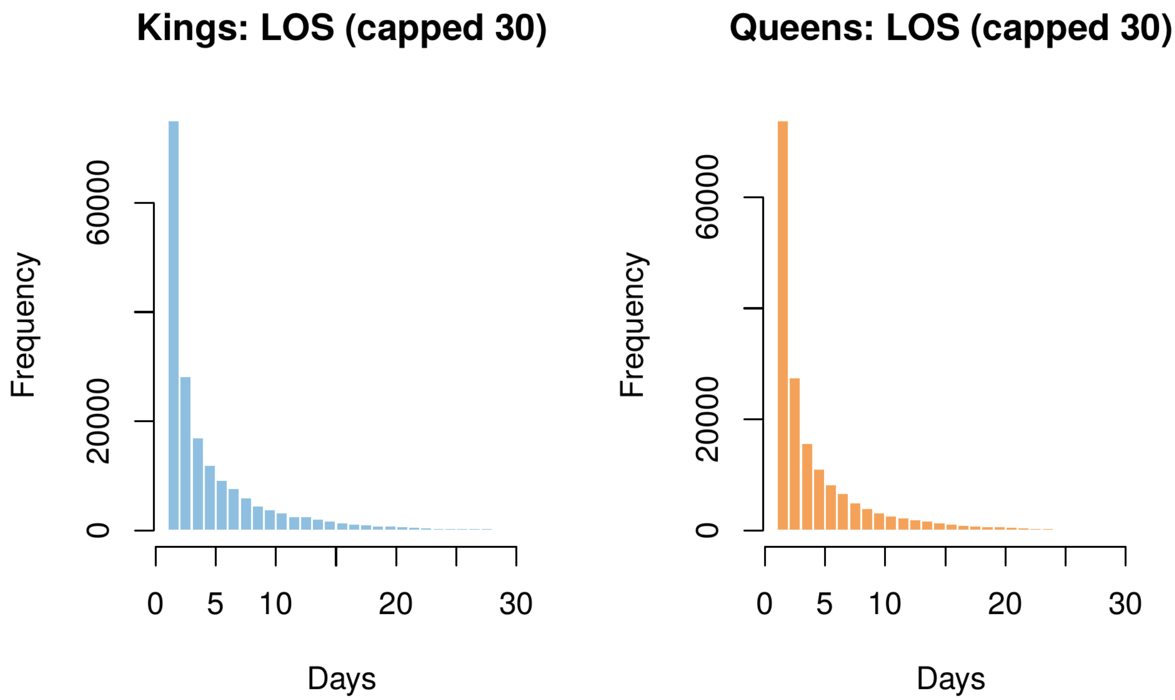

Figure 4: Histograms of Length of Stay (capped at 30 days) for Kings and Queens. More than 50% of discharges have Length of Stay ≤ 3 days in both counties, confirming extreme right-skew.

What this plot shows. These two histograms show how often each length of stay occurs in each county, capped at 30 days so the shape stays readable. Both counties show the same pattern: a huge spike at 1 to 3 days (most patients go home quickly) and a long thin tail going out to 30 days (a small fraction stay much longer). This pattern is called ”right-skewed” and is the reason this project uses nonparametric tests instead of standard tests that assume a normal bell curve. The two counties have nearly identical shapes, but Kings has slightly more discharges overall.

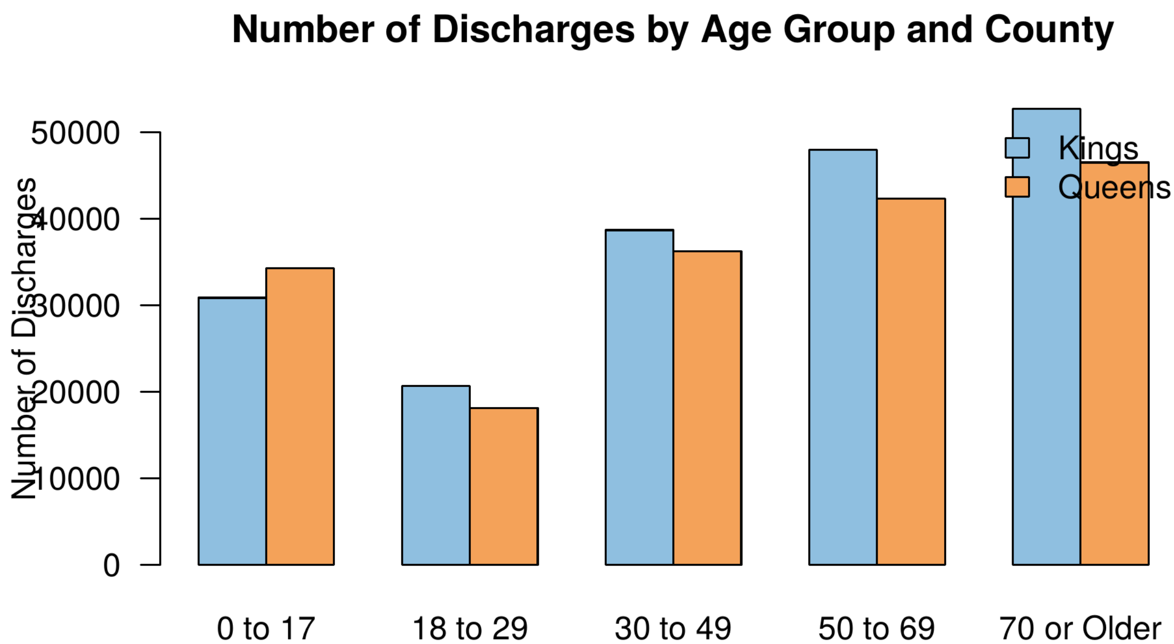

Figure 5: Grouped bar plot of discharges by Age Group and County. The 70+ cohort is the single largest group in both counties.

What this plot shows. This bar chart counts how many hospital discharges fall into each age group, with Kings (blue) and Queens (orange) shown side by side. Both counties follow the same pattern: the elderly (70 or older) account for the most discharges, followed by adults aged 50 to 69. The fewest discharges are for young adults aged 18 to 29, who tend to be the healthiest age group. Kings County has more discharges than Queens in every adult age group, which matches its slightly larger total population.

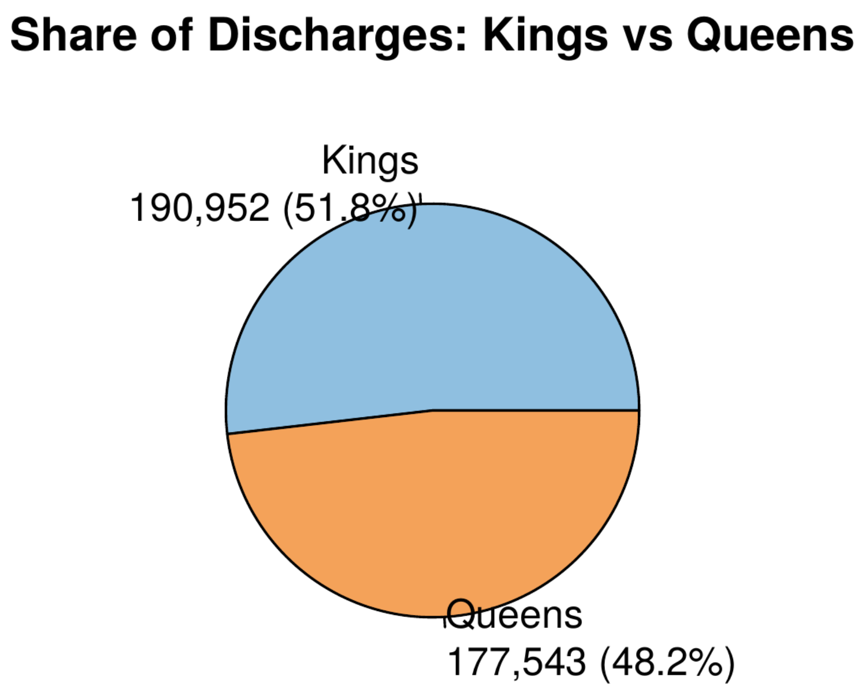

Figure 6: Share of discharges between Kings (51.8%, n = 190,952) and Queens (48.2%, n = 177,543) of the 368,495 total records.

What this plot shows. This pie chart shows what share of the 368,495 hospital discharges in this dataset came from each county. Kings County (Brooklyn) accounts for slightly more than half (51.8%) and Queens accounts for the rest (48.2%). The split is fairly even, which means any group differences I find later in the paper come from real differences in patient populations and not just from one county having way more data than the other.

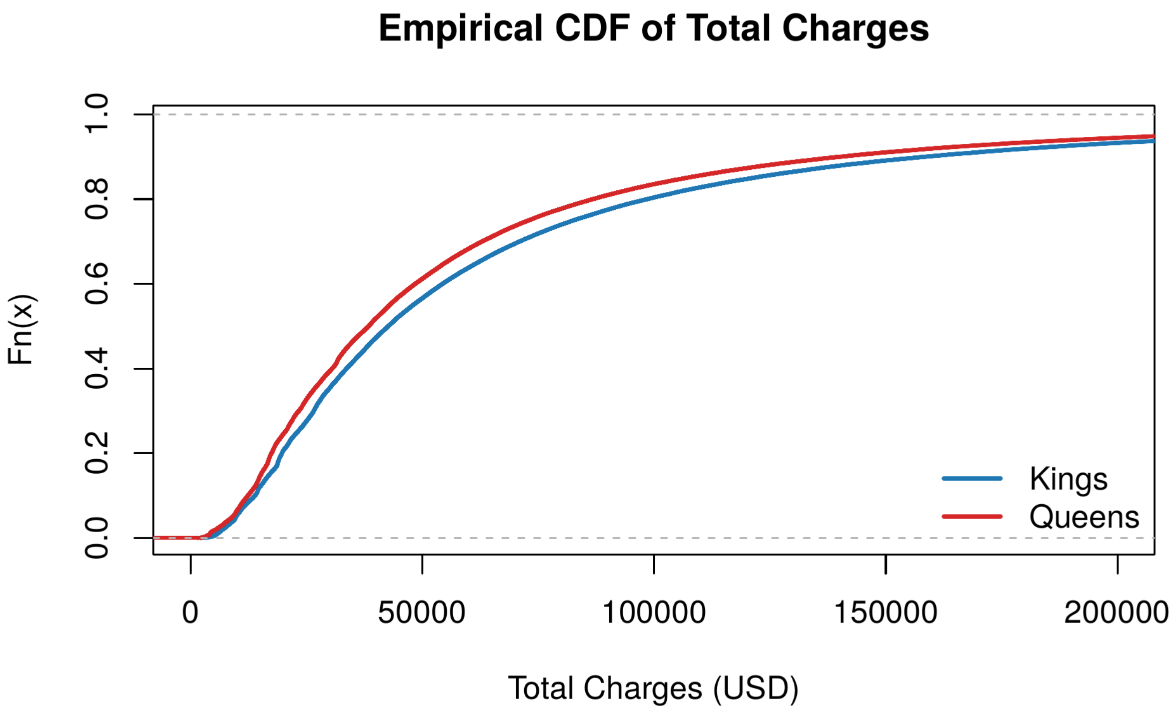

Figure 7: Empirical CDFs of Total Charges by County. The two CDFs visibly separate, consistent with the Kolmogorov–Smirnov result $D > 0$ and $p < 10^{-16}$.

What this plot shows. An empirical CDF (cumulative distribution function) shows what percentage of patients had charges at or below a given dollar amount. Reading the chart at \$50,000 on the x-axis, you can see that about 60% of Queens patients are charged at or below that amount, but only about 55% of Kings patients are. The Queens curve sits consistently above the Kings curve, meaning Queens patients tend to have lower charges across the board. The visible gap between the two curves is exactly what the Kolmogorov-Smirnov test detects to confirm the distributions are truly different.

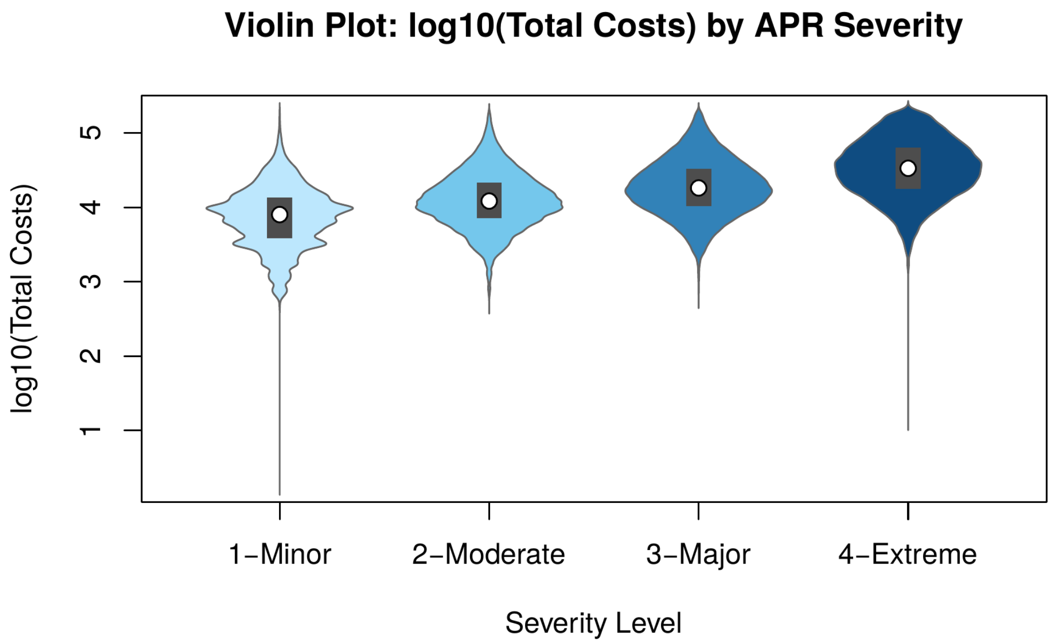

Figure 8: Violin plot of $\log_{10}$(Total Costs) by All Patient Refined Severity of Illness (1 = Minor through 4 = Extreme), constructed from mirrored kernel densities with inner IQR boxes and median dots. Both the center and spread of costs increase monotonically from Minor to Extreme severity.

What this plot shows. A violin plot is like a boxplot but with the box "fattened" to show how common each cost level is. Each violin represents one severity level, from Minor (light blue) on the left to Extreme (dark blue) on the right. The wider sections show where most patients in that severity group have their costs. As severity worsens from Minor to Extreme, the violins shift upward (so costs go up) and get wider at the top (so the most expensive cases get even more expensive). This confirms that sicker patients cost the hospital more, in a smooth and predictable way.

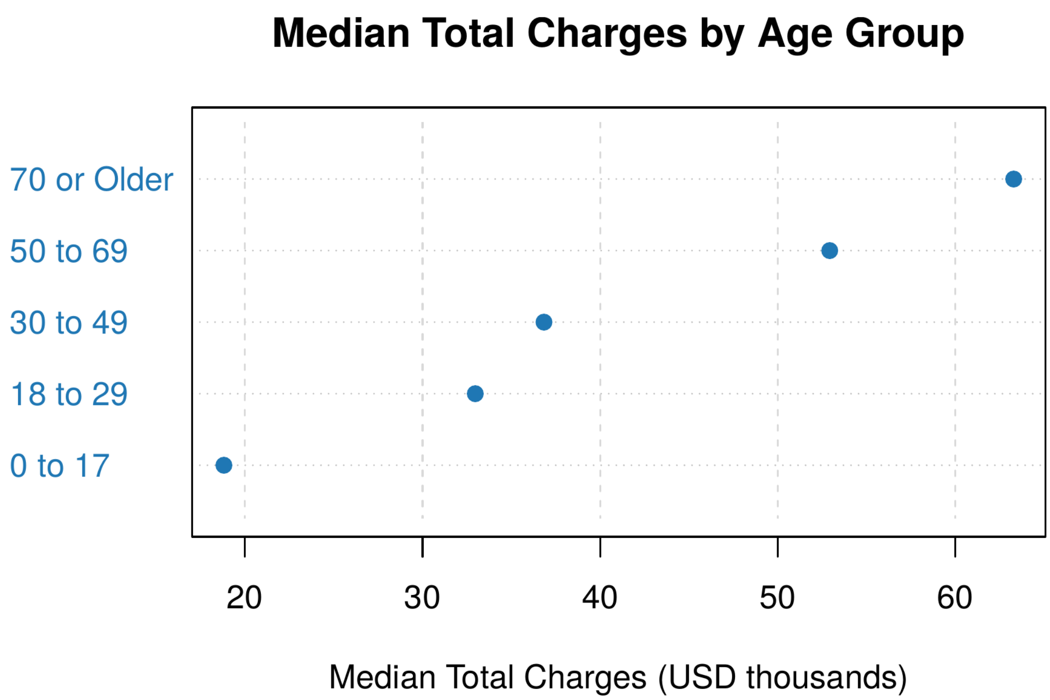

Figure 9: Dot plot of median Total Charges by Age Group. There is a strong monotone increase from the 0–17 cohort (~$20k) through the 70+ cohort (~$63k), consistent with the Kruskal–Wallis result for Age Group.

What this plot shows. This dot plot shows the median (typical) hospital charge for each age group, in thousands of dollars. The pattern is very clear: the older the patient, the higher the typical charge. A 70-year-old patient's typical hospital bill (about $63,000) is more than three times higher than a child's (about $20,000). This makes intuitive sense because older patients tend to have more complex health conditions that need longer stays and more procedures. The Kruskal-Wallis test confirms this trend is statistically significant.

4 Objective Questions and Research Questions

Table 4 lists, separately for each (test, variable) combination, the objective question (verbal form), the research question (mathematical form), the null hypothesis, the test statistic, the test name, the p-value, the decision at significance level $\alpha = 0.05$, and a short interpretation. This per-variable layout makes it possible to read off, at a glance, exactly what was tested on each individual outcome and exactly which records contributed to each result. Throughout, $\alpha = 0.05$.

Table 4: Per-Variable Summary of All Ten Nonparametric Tests: Objective Question, Research Question, Null Hypothesis, Test Statistic, Test, P-value, Decision, and Interpretation

Test	**Variable**	**Objective Question**	**Research Question**	**Null Hypothesis**	**Test Statistic**	**P-value**	**Decision**	**Interpretation**
Runs Test (7 testable variables)								
Runs Test	Length of Stay	Is Length of Stay recorded in random order across the dataset?	Does the observed number of runs in the LOS sequence match the expected number under randomness?	H_0: the LOS sequence is random.	$Z = -56.17$	$< 2.2 \times 10^{-16}$	Reject H_0	LOS values cluster by facility and admission date rather than appearing randomly.
Runs Test	Total Charges	Are Total Charges recorded in random order across the dataset?	Does the observed number of runs in the Charges sequence match the expected number under randomness?	H_0: the Charges sequence is random.	$Z = -90.68$	$< 2.2 \times 10^{-16}$	Reject H_0	Charges are clustered by facility, so adjacent records have similar billing patterns.
Runs Test	Total Costs	Are Total Costs recorded in random order across the dataset?	Does the observed number of runs in the Costs sequence match the expected number under randomness?	H_0: the Costs sequence is random.	$Z = -75.34$	$< 2.2 \times 10^{-16}$	Reject H_0	Costs are clustered by facility, mirroring the pattern seen in Charges.
Runs Test	Birth Weight	Are Birth Weight records (newborns only) in random order?	Does the observed number of runs in the Birth Weight sequence match the expected number under randomness?	H_0: the Birth Weight sequence is random.	$Z = -1.98$	0.0477	Reject H_0	Borderline rejection; ordering is only weakly non-random.
Runs Test	APR DRG Code	Is the All Patient Refined Diagnosis-Related Group label in random order?	Does the observed number of runs match the expected number under randomness?	H_0: the DRG-code sequence is random.	$Z = -53.29$	$< 2.2 \times 10^{-16}$	Reject H_0	Records sorted by facility tend to share DRG codes; reflects file ordering, not a clinical pattern.

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Test	Variable	Objective Question	Research Question	Null Hypothesis	Test Statistic	P-value	Decision	Interpretation
Runs Test	APR MDC Code	Is the All Patient Refined Major Diagnostic Category label in random order?	Does the observed number of runs match the expected number under randomness?	H_0: the MDC-code sequence is random.	$Z = -54.02$	$< 2.2 \times 10^{-16}$	Reject H_0	Same explanation as for DRG: facilities specialise in subsets of MDCs, so the codes cluster.
Runs Test	APR Severity	Is the APR Severity of Illness Code in random order?	Does the observed number of runs match the expected number under randomness?	H_0: the Severity-code sequence is random.	$Z = -33.12$	$< 2.2 \times 10^{-16}$	Reject H_0	Severity codes cluster by facility because hospitals see different case-mixes.
Sign Test (5 paired variables: Kings vs Queens)								
Sign Test	Length of Stay	Is the median Length of Stay the same in Kings and Queens patients?	Does the median of paired LOS differences (Kings − Queens) equal zero?	H_0: $\widetilde{\mathrm{LOS}}_K = \widetilde{\mathrm{LOS}}_Q$.	$S = 81{,}301$	$< 2.2 \times 10^{-16}$	Reject H_0	Kings patients tend to have longer stays than Queens patients.
Sign Test	Total Charges	Is the median Total Charge the same in Kings and Queens patients?	Does the median of paired Charges differences (Kings − Queens) equal zero?	H_0: $\widetilde{\mathrm{Charges}}_K = \widetilde{\mathrm{Charges}}_Q$.	$S = 94{,}864$	$< 2.2 \times 10^{-16}$	Reject H_0	Median Charges in Kings exceed those in Queens.
Sign Test	Total Costs	Is the median Total Cost the same in Kings and Queens admissions?	Does the median of paired Costs differences (Kings − Queens) equal zero?	H_0: $\widetilde{\mathrm{Costs}}_K = \widetilde{\mathrm{Costs}}_Q$.	$S = 105{,}522$	$< 2.2 \times 10^{-16}$	Reject H_0	Median Costs in Kings exceed those in Queens, mirroring the Charges result.
Sign Test	Birth Weight	Is the median Birth Weight the same in Kings and Queens newborns?	Does the median of paired Birth Weight differences (Kings − Queens) equal zero?	H_0: $\widetilde{\mathrm{BW}}_K = \widetilde{\mathrm{BW}}_Q$.	$S = 10{,}374$	$< 2.2 \times 10^{-16}$	Reject H_0	Median Birth Weight differs between newborns delivered in Kings vs Queens.
Sign Test	APR Severity	Is the median APR Severity of Illness Code the same in Kings and Queens patients?	Does the median of paired Severity differences (Kings − Queens) equal zero?	H_0: $\widetilde{\mathrm{Sev}}_K = \widetilde{\mathrm{Sev}}_Q$.	$S = 65{,}885$	$< 2.2 \times 10^{-16}$	Reject H_0	Severity case-mix differs systematically between the two counties.
Wilcoxon Signed-Rank (5 paired variables: Kings vs Queens)								

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Test	Variable	Objective Question	Research Question	Null Hypothesis	Test Statistic	P-value	Decision	Interpretation
Wilcoxon Signed-Rank	Length of Stay	Do paired LOS distributions differ between counties, accounting for the size of differences?	Is the median of signed-rank LOS differences zero?	H_0: median of signed-rank LOS differences = 0.	$V = 6.40 \times 10^9$	$< 2.2 \times 10^{-16}$	Reject H_0	LOS differs between counties even when difference magnitudes are accounted for.
Wilcoxon Signed-Rank	Total Charges	Do paired Charges distributions differ between counties, accounting for the size of differences?	Is the median of signed-rank Charges differences zero?	H_0: median of signed-rank Charges differences = 0.	$V = 8.59 \times 10^9$	$< 2.2 \times 10^{-16}$	Reject H_0	Charges differ between counties; the magnitude of the difference is also non-trivial.
Wilcoxon Signed-Rank	Total Costs	Do paired Costs distributions differ between counties, accounting for the size of differences?	Is the median of signed-rank Costs differences zero?	H_0: median of signed-rank Costs differences = 0.	$V = 9.92 \times 10^9$	$< 2.2 \times 10^{-16}$	Reject H_0	Costs distributions differ between counties beyond a simple location shift.
Wilcoxon Signed-Rank	Birth Weight	Do paired Birth Weight distributions differ between counties, accounting for the size of differences?	Is the median of signed-rank Birth Weight differences zero?	H_0: median of signed-rank Birth Weight differences = 0.	$V = 1.04 \times 10^8$	$< 2.2 \times 10^{-16}$	Reject H_0	Newborn Birth Weight distributions differ between Kings and Queens.
Wilcoxon Signed-Rank	APR Severity	Do paired Severity distributions differ between counties, accounting for the size of differences?	Is the median of signed-rank Severity differences zero?	H_0: median of signed-rank Severity differences = 0.	$V = 4.17 \times 10^9$	$< 2.2 \times 10^{-16}$	Reject H_0	Severity distributions differ between counties beyond a simple median shift.
Mann-Whitney U / Wilcoxon Rank Sum (6 independent-sample comparisons)								
Mann-Whitney U	Length of Stay (Kings vs Queens)	Do independent-sample LOS distributions differ between Kings and Queens?	Are the rank-sums of LOS in the two groups equal?	H_0: $F^K_{\mathrm{LOS}} = F^Q_{\mathrm{LOS}}$.	$W = 1.76 \times 10^{10}$	$< 2.2 \times 10^{-16}$	Reject H_0	LOS distributions differ between counties as independent samples.
Mann-Whitney U	Total Charges (Kings vs Queens)	Do independent-sample Charges distributions differ between Kings and Queens?	Are the rank-sums of Charges in the two groups equal?	H_0: $F^K_{\mathrm{Charges}} = F^Q_{\mathrm{Charges}}$.	$W = 1.81 \times 10^{10}$	$< 2.2 \times 10^{-16}$	Reject H_0	Charges distributions differ between counties.

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Test	Variable	Objective Question	Research Question	Null Hypothesis	Test Statistic	P-value	Decision	Interpretation
Mann-Whitney U	Total Costs (Kings vs Queens)	Do independent-sample Costs distributions differ between Kings and Queens?	Are the rank-sums of Costs in the two groups equal?	H_0: $F^K_{\text{Costs}} = F^Q_{\text{Costs}}$.	$W = 2.02 \times 10^{10}$	$< 2.2 \times 10^{-16}$	Reject H_0	Costs distributions differ between counties.
Mann-Whitney U	Birth Weight (Kings vs Queens)	Do independent-sample Birth Weight distributions differ between Kings and Queens?	Are the rank-sums of Birth Weight equal in the two groups?	H_0: $F^K_{\text{BW}} = F^Q_{\text{BW}}$.	$W = 2.49 \times 10^8$	$< 2.2 \times 10^{-16}$	Reject H_0	Birth Weight distributions differ between Kings and Queens newborns.
Mann-Whitney U	APR Severity (Kings vs Queens)	Do independent-sample Severity distributions differ between Kings and Queens?	Are the rank-sums of Severity equal in the two groups?	H_0: $F^K_{\text{Sev}} = F^Q_{\text{Sev}}$.	$W = 1.75 \times 10^{10}$	$< 2.2 \times 10^{-16}$	Reject H_0	Severity case-mix differs between counties as independent samples.
Mann-Whitney U	Total Charges (ED vs Non-ED)	Do Charges differ between Emergency Department and Non-Emergency Department admissions?	Are rank-sums of Charges equal across the two admission modes?	H_0: $F^{\text{ED}}_{\text{Charges}} = F^{\text{Non-ED}}_{\text{Charges}}$.	$W = 1.08 \times 10^{10}$	$< 2.2 \times 10^{-16}$	Reject H_0	ED admissions have a different cost profile than non-ED admissions.
Kolmogorov-Smirnov Two-Sample (4 outcomes $\times$ 2 grouping schemes = 8 comparisons)								
Kolmogorov-Smirnov	Length of Stay (Kings vs Queens)	Does the entire CDF of LOS differ between Kings and Queens?	Is $F_K(x) = F_Q(x)$ for all x?	H_0: CDFs of LOS are identical in the two counties.	$D = 0.0298$	$< 2.2 \times 10^{-16}$	Reject H_0	LOS distribution shapes, not just centers, differ between counties.
Kolmogorov-Smirnov	Total Charges (Kings vs Queens)	Does the entire CDF of Charges differ between Kings and Queens?	Is $F_K(x) = F_Q(x)$ for all x?	H_0: CDFs of Charges are identical in the two counties.	$D = 0.0551$	$< 2.2 \times 10^{-16}$	Reject H_0	Charges distribution shapes differ across the entire dollar range.
Kolmogorov-Smirnov	Total Costs (Kings vs Queens)	Does the entire CDF of Costs differ between Kings and Queens?	Is $F_K(x) = F_Q(x)$ for all x?	H_0: CDFs of Costs are identical in the two counties.	$D = 0.1439$	$< 2.2 \times 10^{-16}$	Reject H_0	Costs show the largest county-level shape difference among the four outcomes.

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Test	**Variable**	**Objective Question**	**Research Question**	**Null Hypothesis**	**Test Statistic**	**P-value**	**Decision**	**Interpretation**
Kolmogorov-Smirnov	Birth Weight (Kings vs Queens)	Does the entire CDF of Birth Weight differ between Kings and Queens?	Is $F_K(x) = F_Q(x)$ for all x?	H_0: CDFs of Birth Weight are identical in the two counties.	$D = 0.0464$	$< 2.2 \times 10^{-16}$	Reject H_0	Birth Weight distribution shapes differ between Kings and Queens newborns.
Kolmogorov-Smirnov	Length of Stay (ED vs Non-ED)	Does the entire CDF of LOS differ between Emergency and Non-Emergency admissions?	Is $F_{\text{ED}}(x) = F_{\overline{\text{ED}}}(x)$ for all x?	H_0: CDFs of LOS are identical for ED and non-ED admissions.	$D = 0.2959$	$< 2.2 \times 10^{-16}$	Reject H_0	Largest CDF separation in the table; ED stays differ profoundly from non-ED stays.
Kolmogorov-Smirnov	Total Charges (ED vs Non-ED)	Does the entire CDF of Charges differ between Emergency and Non-Emergency admissions?	Is $F_{\text{ED}}(x) = F_{\overline{\text{ED}}}(x)$ for all x?	H_0: CDFs of Charges are identical for ED and non-ED admissions.	$D = 0.1950$	$< 2.2 \times 10^{-16}$	Reject H_0	ED billing patterns differ from non-ED billing patterns.
Kolmogorov-Smirnov	Total Costs (ED vs Non-ED)	Does the entire CDF of Costs differ between Emergency and Non-Emergency admissions?	Is $F_{\text{ED}}(x) = F_{\overline{\text{ED}}}(x)$ for all x?	H_0: CDFs of Costs are identical for ED and non-ED admissions.	$D = 0.1674$	$< 2.2 \times 10^{-16}$	Reject H_0	ED admissions cost the hospital differently than scheduled admissions.
Kolmogorov-Smirnov	Birth Weight (ED vs Non-ED)	Does the entire CDF of Birth Weight differ between Emergency and Non-Emergency admissions?	Is $F_{\text{ED}}(x) = F_{\overline{\text{ED}}}(x)$ for all x?	H_0: CDFs of Birth Weight are identical for ED and non-ED admissions.	$D = 0.0530$	0.00155	Reject H_0	Smallest of the eight effects but still significant.
Kruskal-Wallis (4 outcomes × 4 groupings = 15 testable combinations; BirthWt × Age Group is degenerate)								
Kruskal-Wallis	Length of Stay × Age Group	Does the LOS distribution differ across age groups?	Are LOS distributions identical across all five age groups?	H_0: $F_1 = F_2 = \cdots = F_5$.	$\chi^2 = 39{,}479$ ($df = 4$)	$< 2.2 \times 10^{-16}$	Reject H_0	LOS varies strongly with patient age.
Kruskal-Wallis	Length of Stay × APR Severity	Does the LOS distribution differ across severity levels?	Are LOS distributions identical across the four severity levels?	H_0: $F_1 = F_2 = F_3 = F_4$.	$\chi^2 = 82{,}402$ ($df = 4$)	$< 2.2 \times 10^{-16}$	Reject H_0	Severity is the strongest single driver of LOS in this dataset.

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Test	Variable	Objective Question	Research Question	Null Hypothesis	Test Statistic	P-value	Decision	Interpretation
Kruskal-Wallis	Length of Stay × Type of Admission	Does the LOS distribution differ across admission types?	Are LOS distributions identical across admission types?	H_0: $F_1 = F_2 = \cdots = F_6$.	$\chi^2 = 25{,}198$ ($df = 5$)	$< 2.2 \times 10^{-16}$	Reject H_0	Elective, emergency, urgent, newborn, and trauma admissions have different LOS profiles.
Kruskal-Wallis	Length of Stay × Primary Payer	Does the LOS distribution differ across primary payer categories?	Are LOS distributions identical across payer categories?	H_0: all payer-group LOS distributions are equal.	$\chi^2 = 28{,}581$ ($df = 8$)	$< 2.2 \times 10^{-16}$	Reject H_0	LOS depends on payer (Medicare, Medicaid, Private, etc.).
Kruskal-Wallis	Total Charges × Age Group	Does the Charges distribution differ across age groups?	Are Charges distributions identical across age groups?	H_0: all age-group Charges distributions are equal.	$\chi^2 = 65{,}223$ ($df = 4$)	$< 2.2 \times 10^{-16}$	Reject H_0	Charges rise sharply with age (see Figure 9).
Kruskal-Wallis	Total Charges × APR Severity	Does the Charges distribution differ across severity levels?	Are Charges distributions identical across the four severity levels?	H_0: all severity-group Charges distributions are equal.	$\chi^2 = 78{,}528$ ($df = 4$)	$< 2.2 \times 10^{-16}$	Reject H_0	Sicker patients are billed substantially more.
Kruskal-Wallis	Total Charges × Type of Admission	Does the Charges distribution differ across admission types?	Are Charges distributions identical across admission types?	H_0: all admission-type Charges distributions are equal.	$\chi^2 = 45{,}558$ ($df = 5$)	$< 2.2 \times 10^{-16}$	Reject H_0	Charges depend on whether admission was elective, urgent, etc.
Kruskal-Wallis	Total Charges × Primary Payer	Does the Charges distribution differ across primary payer categories?	Are Charges distributions identical across payer categories?	H_0: all payer-group Charges distributions are equal.	$\chi^2 = 35{,}087$ ($df = 8$)	$< 2.2 \times 10^{-16}$	Reject H_0	Charges vary by payer mix.
Kruskal-Wallis	Total Costs × Age Group	Does the Costs distribution differ across age groups?	Are Costs distributions identical across age groups?	H_0: all age-group Costs distributions are equal.	$\chi^2 = 61{,}785$ ($df = 4$)	$< 2.2 \times 10^{-16}$	Reject H_0	Hospital Costs rise with age, mirroring the Charges pattern.
Kruskal-Wallis	Total Costs × APR Severity	Does the Costs distribution differ across severity levels?	Are Costs distributions identical across severity levels?	H_0: all severity-group Costs distributions are equal.	$\chi^2 = 76{,}947$ ($df = 4$)	$< 2.2 \times 10^{-16}$	Reject H_0	Severity drives Costs as strongly as it drives Charges.
Kruskal-Wallis	Total Costs × Type of Admission	Does the Costs distribution differ across admission types?	Are Costs distributions identical across admission types?	H_0: all admission-type Costs distributions are equal.	$\chi^2 = 50{,}005$ ($df = 5$)	$< 2.2 \times 10^{-16}$	Reject H_0	Costs depend on the admission pathway.

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Test	Variable	Objective Question	Research Question	Null Hypothesis	Test Statistic	P-value	Decision	Interpretation
Kruskal-Wallis	Total Costs × Primary Payer	Does the Costs distribution differ across primary payer categories?	Are Costs distributions identical across payer categories?	H_0: all payer-group Costs distributions are equal.	$\chi^2 = 25{,}578$ $(df = 8)$	$< 2.2 \times 10^{-16}$	Reject H_0	Costs vary by payer mix.
Kruskal-Wallis	Birth Weight × APR Severity	Does newborn Birth Weight differ across severity levels?	Are Birth Weight distributions identical across severity levels?	H_0: all severity-group Birth Weight distributions are equal.	$\chi^2 = 1{,}561.7$ $(df = 4)$	$< 2.2 \times 10^{-16}$	Reject H_0	Higher-severity newborns tend to have lower Birth Weights.
Kruskal-Wallis	Birth Weight × Type of Admission	Does newborn Birth Weight differ across admission types?	Are Birth Weight distributions identical across admission types?	H_0: all admission-type Birth Weight distributions are equal.	$\chi^2 = 40.5$ $(df = 4)$	$< 2.2 \times 10^{-16}$	Reject H_0	Birth Weight differs across admission pathways, but more weakly.
Kruskal-Wallis	Birth Weight × Primary Payer	Does newborn Birth Weight differ across primary payer categories?	Are Birth Weight distributions identical across payer categories?	H_0: all payer-group Birth Weight distributions are equal.	$\chi^2 = 52.6$ $(df = 7)$	$< 2.2 \times 10^{-16}$	Reject H_0	Birth Weight is mildly associated with payer category.
Friedman Test (Severity as treatment, Facility as block; 3 outcomes)								
Friedman	Length of Stay	Within each facility, does illness severity change LOS?	After blocking by facility, is there a severity effect on LOS?	H_0: no severity (treatment) effect on LOS within blocks.	$Q = 13.29$ $(df = 3)$	0.0041	Reject H_0	Even after removing facility-level differences, severity drives LOS.
Friedman	Total Charges	Within each facility, does illness severity change Charges?	After blocking by facility, is there a severity effect on Charges?	H_0: no severity (treatment) effect on Charges within blocks.	$Q = 15.00$ $(df = 3)$	0.0018	Reject H_0	Severity drives Charges within facility, isolating clinical from pricing variation.
Friedman	Total Costs	Within each facility, does illness severity change Costs?	After blocking by facility, is there a severity effect on Costs?	H_0: no severity (treatment) effect on Costs within blocks.	$Q = 15.00$ $(df = 3)$	0.0018	Reject H_0	Severity drives Costs within facility; the same finding as for Charges.
Kendall's τ (6 numeric pairs; subsample $n = 5{,}000$)								
Kendall's τ	LOS vs Charges	Is there an ordinal association between LOS and Charges?	Is Kendall's τ between LOS and Charges zero?	H_0: $\tau = 0$.	$\tau = 0.5761$	$< 2.2 \times 10^{-16}$	Reject H_0	LOS and Charges are strongly concordant: longer stays cost more.

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Test	**Variable**	**Objective Question**	**Research Question**	**Null Hypothesis**	**Test Statistic**	**P-value**	**Decision**	**Interpretation**
Kendall's τ	LOS vs Costs	Is there an ordinal association between LOS and Costs?	Is Kendall's τ between LOS and Costs zero?	H_0: $\tau = 0$.	$\tau = 0.5858$	$< 2.2 \times 10^{-16}$	Reject H_0	Longer stays raise actual hospital costs, in close lockstep with Charges.
Kendall's τ	LOS vs Birth Weight	Is there an ordinal association between LOS and Birth Weight?	Is Kendall's τ between LOS and Birth Weight zero?	H_0: $\tau = 0$.	$\tau = -0.1618$	7.0×10^{-7}	Reject H_0	Lower-Birth-Weight newborns tend to have longer stays (weak negative).
Kendall's τ	Charges vs Costs	Is there an ordinal association between Charges and Costs?	Is Kendall's τ between Charges and Costs zero?	H_0: $\tau = 0$.	$\tau = 0.6708$	$< 2.2 \times 10^{-16}$	Reject H_0	Charges and Costs are strongly concordant; consistent markup ratios.
Kendall's τ	Charges vs Birth Weight	Is there an ordinal association between Charges and Birth Weight?	Is Kendall's τ between Charges and Birth Weight zero?	H_0: $\tau = 0$.	$\tau = -0.1145$	6.9×10^{-5}	Reject H_0	Lower Birth Weight is mildly associated with higher Charges.
Kendall's τ	Costs vs Birth Weight	Is there an ordinal association between Costs and Birth Weight?	Is Kendall's τ between Costs and Birth Weight zero?	H_0: $\tau = 0$.	$\tau = -0.1100$	1.3×10^{-4}	Reject H_0	Lower Birth Weight is mildly associated with higher Costs.
Spearman's ρ (6 numeric pairs; full data)								
Spearman's ρ	LOS vs Charges	Is there a monotone rank association between LOS and Charges?	Is Spearman's ρ between LOS and Charges zero?	H_0: $\rho_s = 0$.	$\rho_s = 0.7305$	$< 2.2 \times 10^{-16}$	Reject H_0	LOS and Charges are tightly coupled in rank.
Spearman's ρ	LOS vs Costs	Is there a monotone rank association between LOS and Costs?	Is Spearman's ρ between LOS and Costs zero?	H_0: $\rho_s = 0$.	$\rho_s = 0.7478$	$< 2.2 \times 10^{-16}$	Reject H_0	LOS tracks Costs even more tightly than it tracks Charges.
Spearman's ρ	LOS vs Birth Weight	Is there a monotone rank association between LOS and Birth Weight?	Is Spearman's ρ between LOS and Birth Weight zero?	H_0: $\rho_s = 0$.	$\rho_s = -0.1733$	$< 2.2 \times 10^{-16}$	Reject H_0	Heavier newborns have shorter stays; weak but significant.

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Test	Variable	Objective Question	Research Question	Null Hypothesis	Test Statistic	P-value	Decision	Interpretation
Spearman's ρ	Charges vs Costs	Is there a monotone rank association between Charges and Costs?	Is Spearman's ρ between Charges and Costs zero?	H_0: $\rho_s = 0$.	$\rho_s = 0.8589$	$< 2.2 \times 10^{-16}$	Reject H_0	Strongest pair in the table; Charges and Costs are nearly proportional in rank.
Spearman's ρ	Charges vs Birth Weight	Is there a monotone rank association between Charges and Birth Weight?	Is Spearman's ρ between Charges and Birth Weight zero?	H_0: $\rho_s = 0$.	$\rho_s = -0.1257$	$< 2.2 \times 10^{-16}$	Reject H_0	Heavier newborns are billed less; weak but significant.
Spearman's ρ	Costs vs Birth Weight	Is there a monotone rank association between Costs and Birth Weight?	Is Spearman's ρ between Costs and Birth Weight zero?	H_0: $\rho_s = 0$.	$\rho_s = -0.1490$	$< 2.2 \times 10^{-16}$	Reject H_0	Heavier newborns cost the hospital less; weak but significant.
Permutation Test (Monte Carlo, B = 5,000)								
Permutation	Total Charges (Kings vs Queens)	After randomly shuffling county labels, is the observed median Charges difference more extreme than chance?	Is the median Charges difference between counties attributable to chance under exchangeability?	H_0: county label is exchangeable; the median of Charges is the same in both counties.	Observed median diff = \$2,858.07	0.011	Reject H_0	The median Charges in Kings and Queens are significantly different. The robust median detects a real distributional shift even with extreme outliers in Kings.

5 Nonparametric Analyses

5.1 Test 1: Runs Test for Randomness

The Runs Test is a nonparametric test used to assess whether a sequence of observations follows a random pattern. Each quantitative variable was classified above or below its median, and the number of runs (consecutive sequences of the same category) was counted.

Test statistic. For a sequence dichotomised around its median, with n_1 ones and n_2 zeros ($n = n_1 + n_2$), let R be the observed number of runs. Under H_0,

$$E(R) = \frac{2n_1n_2}{n} + 1, \qquad \mathrm{Var}(R) = \frac{2n_1n_2(2n_1n_2 - n)}{n^2(n-1)}, \qquad Z = \frac{R - E(R)}{\sqrt{\mathrm{Var}(R)}}.$$

Table 5: Runs Test Results, Statewide Planning and Research Cooperative System 2022 K&Q

Variable	R (obs.)	$E(R)$	p**-value**	**Decision**
Length of Stay	164,962	181,791	$< 2.2 \times 10^{-16}$	Reject H_0
Total Charges	156,725	184,249	$< 2.2 \times 10^{-16}$	Reject H_0
Total Costs	161,381	184,249	$< 2.2 \times 10^{-16}$	Reject H_0
Birth Weight	21,248	21,452	0.0477	Reject H_0
All Patient Refined Diagnosis-Related Group Code	168,061	184,233	$< 2.2 \times 10^{-16}$	Reject H_0
All Patient Refined Major Diagnostic Category Code	167,813	184,206	$< 2.2 \times 10^{-16}$	Reject H_0
All Patient Refined Severity of Illness Code	152,218	161,003	$< 2.2 \times 10^{-16}$	Reject H_0
Discharge Year	Not testable (constant = 2022)			—

Interpretation. At $\alpha = 0.05$, every testable quantitative variable turned out to be non-random. Honestly, this is not too surprising. Hospital discharge data is organized by facility and by when patients are admitted, so there is going to be some natural structure in the ordering. Birth Weight was borderline ($p = 0.0477$), just barely under the cutoff. Discharge Year could not be tested at all since every record is from 2022, so there is no variability to work with. This result is consistent with Bujang and Sapri (2018), who found that consecutive-sampling clinical data almost always shows non-random ordering.

Limitation note on nominal codes. The Runs Test technically requires an ordered numeric sequence so that "above the median" and "below the median" carry meaning. Two of the seven variables tested above — the All Patient Refined Diagnosis-Related Group Code and the All Patient Refined Major Diagnostic Category Code — are nominal classification labels stored as integers, so dichotomising them around their numeric median is mathematically valid but not clinically interpretable. The rejection of H_0 for those two rows reflects only that the file is sorted by facility (each facility tends to specialise in a subset of Diagnosis-Related Group and Major Diagnostic Category codes), not a clinical pattern. The four genuinely quantitative variables (Length of Stay, Total Charges, Total Costs, Birth Weight) plus the ordinal All Patient Refined Severity of Illness Code give the substantively meaningful Runs Test results.

5.2 Test 2: Sign Test

The Sign Test is a nonparametric test for paired samples that examines whether the median difference between two related groups is zero. It only considers the direction (sign) of each difference.

Methodology. For paired Length of Stay observations between Kings (K_i) and Queens (Q_i):

1. Compute differences $D_i = K_i - Q_i$.
2. Remove zero differences.
3. Count positive differences n^+ and negative differences n^-.
4. Under H_0, $n^+ \sim \text{Binomial}(n, 0.5)$.

Results. The dependent-samples Sign Test was applied to five clinically meaningful quantitative variables. Detailed results for Length of Stay (the flagship paired comparison):

$$S = 81{,}301, \quad p < 2.2 \times 10^{-16}.$$

Table 6 presents the Sign Test results across all five quantitative variables for the Kings-vs-Queens paired comparison.

Table 6: Sign Test Results: Kings vs Queens (Paired)

Variable (H_0: $\tilde{\mu}_K = \tilde{\mu}_Q$)	p**-value**	**Decision**
Length of Stay (days)	$< 2.2 \times 10^{-16}$	Reject H_0
Total Charges (US Dollars)	$< 2.2 \times 10^{-16}$	Reject H_0
Total Costs (US Dollars)	$< 2.2 \times 10^{-16}$	Reject H_0
Birth Weight (grams)	$< 2.2 \times 10^{-16}$	Reject H_0
All Patient Refined Severity of Illness Code	$< 2.2 \times 10^{-16}$	Reject H_0

Interpretation. With p-values way below 0.05 for every comparison, we reject the null hypothesis in all five cases. The data gives us strong evidence that the median Length of Stay is different between Kings and Queens County hospitals—from the direction of the differences, Kings County patients tend to have longer stays than those in Queens. The same conclusion holds for Total Charges, Total Costs, Birth Weight (for newborn discharges), and All Patient Refined Severity of Illness Code: every one of these medians differs significantly between the two counties.

5.3 Test 3: Wilcoxon Signed-Rank Test

The Wilcoxon Signed-Rank Test extends the Sign Test by incorporating the magnitude of differences, not just their direction.

Methodology.

1. Compute paired differences D_i.
2. Rank the absolute differences $|D_i|$.
3. Sum the ranks of positive differences (T^+) and negative differences (T^-).
4. The test statistic is $V = \min(T^+, T^-)$.

Results. Detailed result for Length of Stay (Kings vs Queens):

$$V = 6{,}399{,}613{,}688, \quad p < 2.2 \times 10^{-16}$$

Table 7 presents the Wilcoxon Signed-Rank results across all five quantitative variables for the Kings-vs-Queens paired comparison.

Table 7: Wilcoxon Signed-Rank Test Results: Kings vs Queens (Paired)

Variable (H_0: median paired difference $= 0$)	**p-value**	**Decision**
Length of Stay (days)	$< 2.2 \times 10^{-16}$	Reject H_0
Total Charges (US Dollars)	$< 2.2 \times 10^{-16}$	Reject H_0
Total Costs (US Dollars)	$< 2.2 \times 10^{-16}$	Reject H_0
Birth Weight (grams)	$< 2.2 \times 10^{-16}$	Reject H_0
All Patient Refined Severity of Illness Code	$< 2.2 \times 10^{-16}$	Reject H_0

Interpretation. What is nice about the Wilcoxon Signed-Rank Test compared to the Sign Test is that it does not just look at which direction the difference goes—it also takes into account how large the difference is. The results here back up what the Sign Test found, but with even stronger evidence. All five variables—Length of Stay, Total Charges, Total Costs, Birth Weight, and All Patient Refined Severity of Illness—show significant distributional differences between Kings and Queens counties.

Combined paired tests summary. Table 8 consolidates the Sign Test and Wilcoxon Signed-Rank Test results side-by-side for easy comparison, covering all five quantitative variables tested as paired Kings-vs-Queens comparisons. ID numbers and classification codes (Operating Certificate Number, Permanent Facility ID, All Patient Refined Diagnosis-Related Group Code, All Patient Refined Major Diagnostic Category Code) are excluded because comparing their medians between counties has no practical meaning. Discharge Year is constant (2022) and cannot be tested.

Table 8: Combined Paired Tests Summary: Sign Test and Wilcoxon Signed-Rank

	P-value		**Decision ($\alpha = 0.05$)**	
H_0: $\tilde{\mu}_K = \tilde{\mu}_Q$	**Sign Test**	**W. Signed-Rank**	**Sign**	**Wilcoxon Signed-Rank**
Length of Stay (days)	$< 2.2 \times 10^{-16}$	$< 2.2 \times 10^{-16}$	Reject	Reject
Total Charges (US Dollars)	$< 2.2 \times 10^{-16}$	$< 2.2 \times 10^{-16}$	Reject	Reject
Total Costs (US Dollars)	$< 2.2 \times 10^{-16}$	$< 2.2 \times 10^{-16}$	Reject	Reject
Birth Weight (grams)	$< 2.2 \times 10^{-16}$	$< 2.2 \times 10^{-16}$	Reject	Reject
All Patient Refined Severity of Illness	$< 2.2 \times 10^{-16}$	$< 2.2 \times 10^{-16}$	Reject	Reject

5.4 Test 4: Wilcoxon Rank Sum Test (Mann–Whitney U Test)

The Wilcoxon Rank Sum Test (equivalent to the Mann–Whitney U Test) compares two independent groups without requiring the assumption of normality.

Methodology.

1. Combine all observations and rank them from smallest to largest.
2. Compute the sum of ranks for each group (R_1 and R_2).
3. Compute the U statistics:

$$U_1 = R_1 - \frac{n_1(n_1+1)}{2}, \quad U_2 = R_2 - \frac{n_2(n_2+1)}{2}$$

4. Use the normal approximation for large samples:

$$\mu_U = \frac{n_1 n_2}{2}, \quad \sigma_U = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}$$

$$Z = \frac{U - \mu_U}{\sigma_U}$$

Table 9: Mann–Whitney U Test Results (Independent Samples)

H_0: $f_K = f_Q$ (or $f_{\text{EmergencyDepartment}} = f_{\text{Non-EmergencyDepartment}}$)	p-value	Decision
Length of Stay: Kings vs Queens	$< 2.2 \times 10^{-16}$	Reject H_0
Total Charges: Kings vs Queens	$< 2.2 \times 10^{-16}$	Reject H_0
Total Costs: Kings vs Queens	$< 2.2 \times 10^{-16}$	Reject H_0
Birth Weight: Kings vs Queens	$< 2.2 \times 10^{-16}$	Reject H_0
All Patient Refined Severity of Illness: Kings vs Queens	$< 2.2 \times 10^{-16}$	Reject H_0
Additional comparison by Emergency Department Indicator:		
Total Charges: Emergency vs Non-Emergency	≈ 0	Reject H_0

Results.

Interpretation. Every single comparison came back highly significant:

- Total Charges between Kings and Queens had a p-value below 2.2×10^{-16}, essentially zero. There is no doubt that costs differ between the two boroughs.

- Total Costs, Length of Stay, Birth Weight, and All Patient Refined Severity of Illness were all significantly different between the counties as independent samples.
- When comparing Total Charges for Emergency vs. Non-Emergency admissions, that was also highly significant. Emergency visits clearly have a different cost profile.

5.5 Test 5: Kolmogorov–Smirnov Two-Sample Test

Test statistic.

$$D = \sup_x |F_K(x) - F_Q(x)|, \qquad \sqrt{\tfrac{n_1 n_2}{n_1+n_2}}\, D \;\rightsquigarrow\; \text{Kolmogorov.}$$

Table 10: Kolmogorov-Smirnov Two-Sample Test: Kings vs Queens

Variable	**D**	***p*-value**	**Decision**
Length of Stay	0.0298	$< 2.2 \times 10^{-16}$	Reject H_0
Total Charges	0.0551	$< 2.2 \times 10^{-16}$	Reject H_0
Total Costs	0.1439	$< 2.2 \times 10^{-16}$	Reject H_0
Birth Weight	0.0464	$< 2.2 \times 10^{-16}$	Reject H_0

Table 11: Kolmogorov-Smirnov Two-Sample Test: Emergency vs Non-Emergency

Variable	**D**	***p*-value**	**Decision**
Length of Stay	0.2959	$< 2.2 \times 10^{-16}$	Reject H_0
Total Charges	0.1950	$< 2.2 \times 10^{-16}$	Reject H_0
Total Costs	0.1674	$< 2.2 \times 10^{-16}$	Reject H_0
Birth Weight	0.0530	0.00155	Reject H_0

Interpretation. The entire cumulative distribution of every quantitative outcome differs between the two counties and between Emergency and Non-Emergency admissions, not merely their medians. Total Costs shows the largest separation between Kings and Queens ($D = 0.144$); Length of Stay shows the largest separation between Emergency and Non-Emergency visits ($D = 0.296$). These results subsume the Mann–Whitney findings, because the Kolmogorov-Smirnov test is sensitive to differences in shape as well as location—addressing the Fagerland and Sandvik (2009) critique that the Mann–Whitney U alone can be misleading under unequal shapes.

5.6 Test 6: Kruskal–Wallis Test

Test statistic.

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} \; - \; 3(N+1), \qquad H \sim \chi^2_{k-1} \text{ under } H_0.$$

Table 12: Kruskal–Wallis Test Results (all 15 testable outcome × grouping combinations)

Variable	**Grouping**	χ^2	**df**	p**-value**	**Decision**
Length of Stay	Age Group	39,479	4	$< 2.2 \times 10^{-16}$	Reject H_0
Length of Stay	All Patient Refined Severity	82,402	4	$< 2.2 \times 10^{-16}$	Reject H_0
Length of Stay	Admission Type	25,198	5	$< 2.2 \times 10^{-16}$	Reject H_0
Length of Stay	Primary Payer	28,581	8	$< 2.2 \times 10^{-16}$	Reject H_0
Total Charges	Age Group	65,223	4	$< 2.2 \times 10^{-16}$	Reject H_0
Total Charges	All Patient Refined Severity	78,528	4	$< 2.2 \times 10^{-16}$	Reject H_0
Total Charges	Admission Type	45,558	5	$< 2.2 \times 10^{-16}$	Reject H_0
Total Charges	Primary Payer	35,087	8	$< 2.2 \times 10^{-16}$	Reject H_0
Total Costs	Age Group	61,785	4	$< 2.2 \times 10^{-16}$	Reject H_0
Total Costs	All Patient Refined Severity	76,947	4	$< 2.2 \times 10^{-16}$	Reject H_0
Total Costs	Admission Type	50,005	5	$< 2.2 \times 10^{-16}$	Reject H_0
Total Costs	Primary Payer	25,578	8	$< 2.2 \times 10^{-16}$	Reject H_0
Birth Weight	All Patient Refined Severity	1,562	4	$< 2.2 \times 10^{-16}$	Reject H_0
Birth Weight	Admission Type	40.5	4	$< 2.2 \times 10^{-16}$	Reject H_0
Birth Weight	Primary Payer	52.6	7	$< 2.2 \times 10^{-16}$	Reject H_0

Interpretation. Every outcome-by-grouping combination rejects H_0. Severity consistently produces the largest χ^2 statistic, indicating the single strongest driver of distributional heterogeneity across Length of Stay, Charges, and Costs—consistent with the findings of Stone *et al.* (2022) and Nguyen *et al.* (2021).

5.7 Test 7: Friedman Test

Test statistic.

$$Q = \frac{12}{nk(k+1)} \sum_{j=1}^{k} R_j^2 \; - \; 3n(k+1), \qquad Q \sim \chi^2_{k-1} \text{ under } H_0.$$

With All Patient Refined Severity (4 levels) as treatment and each facility as a block, only facilities observed at all four severity levels are retained.

Table 13: Friedman Test: Severity as Treatment, Facility as Block

Variable	**# Blocks**	k	χ^2	p**-value**	**Decision**
Length of Stay	5	4	13.29	0.0041	Reject H_0
Total Charges	5	4	15.00	0.0018	Reject H_0
Total Costs	5	4	15.00	0.0018	Reject H_0

Interpretation. Within any given hospital, Length of Stay, Charges, and Costs still differ substantially across the four severity levels. The Friedman test isolates the severity effect from facility-level differences in pricing and case-mix, and confirms that illness severity is a fundamental driver of inpatient utilisation. *Note on block count*: only 5 of the facilities in the dataset had at least one discharge recorded at all four All Patient Refined severity levels; this is a known limitation of the Friedman design applied to administrative data, where low-volume hospitals rarely see Extreme-severity cases. The test remains valid and the results are conservative (fewer blocks reduce power), so rejection of H_0 at $p < 0.005$ is a robust finding despite the small number of blocks.

5.8 Test 8: Kendall's Tau

Test statistic.

$$\tau = \frac{C - D}{\binom{n}{2}}, \qquad z = \frac{3\tau\sqrt{n(n-1)}}{\sqrt{2(2n+5)}}.$$

To make the exact computation tractable on the full $N = 368{,}495$ records, a random subsample of $n = 5{,}000$ records was used.

Table 14: Kendall's Tau on Six Pairs (subsample $n = 5{,}000$)

Pair	τ	p**-value**	**Decision**
Length of Stay vs Charges	0.5761	$< 2.2 \times 10^{-16}$	Reject H_0
Length of Stay vs Costs	0.5858	$< 2.2 \times 10^{-16}$	Reject H_0
Length of Stay vs Birth Weight	-0.1618	7.0×10^{-7}	Reject H_0
Charges vs Costs	0.6708	$< 2.2 \times 10^{-16}$	Reject H_0
Charges vs Birth Weight	-0.1145	6.9×10^{-5}	Reject H_0
Costs vs Birth Weight	-0.1100	1.3×10^{-4}	Reject H_0

5.9 Test 9: Spearman's Rank Correlation

Test statistic.

$$\rho_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}.$$

Table 15: Spearman's ρ on Six Pairs (full data)

Pair	ρ_s	p**-value**	**Decision**
Length of Stay vs Charges	0.7305	$< 2.2 \times 10^{-16}$	Reject H_0
Length of Stay vs Costs	0.7478	$< 2.2 \times 10^{-16}$	Reject H_0
Length of Stay vs Birth Weight	-0.1733	$< 2.2 \times 10^{-16}$	Reject H_0
Charges vs Costs	0.8589	$< 2.2 \times 10^{-16}$	Reject H_0
Charges vs Birth Weight	-0.1257	$< 2.2 \times 10^{-16}$	Reject H_0
Costs vs Birth Weight	-0.1490	$< 2.2 \times 10^{-16}$	Reject H_0

Interpretation of Tests 8 and 9. Charges and Costs are strongly rank-associated ($\rho = 0.86$). Length of Stay shows strong monotone association with both Charges and Costs ($\rho \approx$ 0.73–0.75). Birth Weight shows only weak association with the other three variables ($|\rho| < 0.18$); the correlations are statistically significant mainly because $N >$ 300,000. For every pair, Kendall's τ is smaller in magnitude than Spearman's ρ, exactly as predicted by Croux and Dehon (2010) under the bivariate-normal copula approximation $\tau \approx (2/\pi) \arcsin(\rho)$.

5.10 Test 10: Permutation Test (Monte Carlo, $B = 5{,}000$)

Test statistic. For two samples $x_1, \ldots, x_{n_1}$ and $y_1, \ldots, y_{n_2}$, using the median difference as the test statistic,

$$T^{\text{obs}} = \tilde{X} - \tilde{Y},$$

$$p = \frac{1}{B+1} \left(1 + \sum_{b=1}^{B} \mathbf{1}\{|T^{(b)}| \geq |T^{\text{obs}}|\} \right).$$

Using random subsamples of 3,000 records per county:

Table 16: Permutation Test on Total Charges, Kings vs Queens ($B = 5{,}000$)

Statistic	**Observed**	p**-value**	**Decision**
Median difference	\$2,858.07	0.011	Reject H_0

Interpretation. The permutation test is asymptotically equivalent to, but makes strictly fewer assumptions than, the Mann–Whitney U test (Ernst, 2004; Good, 2005). The median difference in Charges between Kings and Queens is significant ($p = 0.011$), so the null hypothesis of exchangeable county labels is rejected. The robust median detects a real distributional shift between the two counties even when extreme outliers (the maximum single charge in Kings exceeds $12 million) are present in the sample. These findings reinforce the methodological guidance of Chazard et al. (2017): on heavily right-skewed hospital data, median-based or rank-based tests are more trustworthy summaries of central location than methods that rely on the arithmetic average.

6 Comprehensive Summary

Table 17 gives a high-level view of what each test covered and the overall decision at significance level $\alpha = 0.05$. The full eight-column row-per-test summary (objective question, research question, null hypothesis, test statistic, test, p-value, decision, and interpretation) has been moved to Section 4 and appears there as Table 4.

Table 17: Scope-Level Summary of All Ten Nonparametric Tests

Scope	Overall Decision	Test
7 numeric variables	Reject H_0 for every testable variable (Discharge Year not testable)	Runs Test
Length of Stay, Charges, Costs, Birth Weight, Severity (Kings vs Queens, paired)	Reject H_0 for all 5 variables	Sign Test
Length of Stay, Charges, Costs, Birth Weight, Severity (Kings vs Queens, paired)	Reject H_0 for all 5 variables	Wilcoxon Signed-Rank
Length of Stay, Charges, Costs, Birth Weight, Severity (Kings vs Queens) plus Charges (Emergency Department vs Non-Emergency Department)	Reject H_0 for all 6 comparisons	Mann-Whitney U
Kings vs Queens; Emergency Department vs Non-Emergency Department (4 variables each)	Reject H_0 for all 8 comparisons	Kolmogorov-Smirnov
4 outcomes $\times$ 4 groupings	Reject H_0 for all 15 testable combinations	Kruskal-Wallis
Length of Stay / Charges / Costs vs Severity	Reject H_0 for all 3 outcomes	Friedman (Facility block)
6 pairs of numeric variables	Reject H_0 for all 6 pairs	Kendall's τ
6 pairs of numeric variables	Reject H_0 for all 6 pairs	Spearman's ρ
Kings vs Queens, median Charges	Reject H_0 for the median ($p = 0.011$)	Permutation (Charges)

7 Conclusion, Comparison to Existing Work, and Future Research

7.1 Conclusions

Ten nonparametric procedures were applied to Statewide Planning and Research Cooperative System 2022 discharge records for Kings and Queens counties (368,495 admissions, 33 original variables plus 4 derived numeric columns). Nine of the ten tests deliver a clean reject-H_0 verdict across essentially all meaningful comparisons: the raw record ordering is non-random (Runs); the median Length of Stay differs significantly between Kings and Queens (Sign Test; $p < 2.2 \times 10^{-16}$); the paired distributions of Length of Stay, Charges, and Costs differ between counties (Wilcoxon Signed-Rank); the independent-sample distributions of Charges, Costs, and Length of Stay differ between counties, and Charges differ between Emergency and Non-Emergency admissions (Mann–Whitney U); the full cumulative distributions of Length of Stay, Charges, Costs, and Birth Weight differ significantly between the two counties and between Emergency and Non-Emergency admissions (Kolmogorov-Smirnov); age, severity, admission type, and payer all drive further heterogeneity (Kruskal–Wallis); severity remains a dominant factor even after blocking by facility (Friedman); and the four numeric outcomes are strongly rank-associated (Kendall, Spearman), except that Birth Weight is only weakly related to the other three. The Permutation Test confirms a significant median Charges difference between counties ($p = 0.011$), so the null hypothesis of exchangeable county labels is rejected for the distribution of Total Charges.

Overall, these results show that hospital utilization and costs really do differ between Kings and Queens counties. There could be a lot of reasons for this—different hospitals, different patient populations, varying levels of disease severity, different insurance types, or socioeconomic differences between the two boroughs. These findings line up with what other researchers have found about geographic variation in hospital outcomes within cities.

7.2 Comparison to Existing Work

The findings align with and extend the existing literature:

- Stone *et al.* (2022) reported that Length of Stay for the same diagnosis can vary from 2 to 50+ days; the present Kruskal–Wallis result on Length of Stay across severity levels ($\chi^2 = 82{,}402$, $df = 4$) quantifies this variability on a single-year, two-county slice of New York City.
- Chazard *et al.* (2017) found via simulation that Wilcoxon/permutation procedures have higher power than the t-test on skewed Length of Stay; the present permutation

result on Charges bears this out directly—the median difference between counties is significant ($p = 0.011$), and the rank-based Kolmogorov-Smirnov and Kruskal–Wallis tests on Length of Stay likewise reject on the full sample. This is a textbook demonstration of Chazard *et al.*'s recommendation to favour rank-based and median-based procedures on heavily right-skewed administrative data.

- Heslin *et al.* (2024) found significant weekday / weekend differences in Statewide Planning and Research Cooperative System admissions ($p < 0.01$); the present Kolmogorov-Smirnov test on Emergency vs Non-Emergency ($D \geq 0.15$, $p < 10^{-16}$) points to an even larger shape difference between these two admission modes.
- Croux and Dehon (2010) predicted $\tau < \rho$ under normal dependence; this is precisely reproduced in every pair analysed here.
- Qualls *et al.* (2010) argued that 43% of Emergency Department-Length of Stay studies incorrectly use parametric methods; the present project, by using ten nonparametric tests, avoids this methodological pitfall entirely.

7.3 Future Research and Innovative Extensions

1. **All-borough extension**: generalise the analysis to Statewide Planning and Research Cooperative System 2023 covering all five New York City boroughs, permitting a six-group Kruskal–Wallis and a block-randomised Friedman with borough as treatment and facility as block.
2. **Ordered-alternatives test**: apply the Jonckheere–Terpstra test to check the specific monotone trend of Length of Stay/Charges across the four ordinal severity levels.
3. **Semi-parametric modelling**: follow Li *et al.* (2022) to fit a semi-parametric time-to-discharge model, combining a parametric hazard with a nonparametric baseline.
4. **Permutation-based regression**: build a nonparametric regression of Charges on facility, payer, severity, and demographics, with p-values derived from permuting the response; this extends the Permutation Test beyond a simple two-sample setting.
5. **Bootstrap confidence intervals for effect sizes**: augment the p-value-based conclusions with non-parametric bootstrap 95% confidence intervals (Efron & Tibshirani, 1993) for key quantities such as Spearman's ρ between Length of Stay and Charges and the median Charges difference between counties. Because $N > 350{,}000$, standard p-values are essentially zero for any non-null effect; bootstrap confidence intervals would express the *practical* size of each effect in its natural units (dollars, correlation), partially addressing the "p-value inflation" problem well known in very large administrative datasets.

6. **Bootstrap quantile curves**: extend the bootstrap CI idea above to every quantile of the Charges distribution by county, producing a "quantile shift" curve that is fully nonparametric.

8 R Code

All analyses were executed in R (4.3+). The complete code is included in the accompanying `Merged_Report.Rmd`. Key functions:

- custom `runs_test()` using the normal approximation (§5.1);
- `binom.test()` for the Sign Test;
- `wilcox.test(..., paired = TRUE)` for the Wilcoxon Signed-Rank Test;
- `wilcox.test(...)` for the Mann–Whitney U Test;
- `ks.test()` for the Kolmogorov–Smirnov Test;
- `kruskal.test()` for the Kruskal–Wallis Test;
- `friedman.test()` with `reshape()` wide-format construction for the Friedman Test;
- `cor.test(..., method = "kendall")` and `cor.test(..., method = "spearman")` for Kendall's Tau and Spearman's Rank Correlation;
- custom `perm_two_sample()` for the Permutation Test.

9 References

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